Machine Learning to Predict the Freeway Traffic Accidents-Based Driving Simulation

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***Abstract*—The aimed of this research is to evaluate and compare different approaches to modeling crash severity as well as investigating the effect of risk factors on the fatality outcomes of traffics crashes using machine learning-based driving simulation. We developed prediction models to identify risk factors of traffics crashes can be targeted to reduce accident. The Random Forest model demonstrated the best performance from among the six different techniques with accurate 82.6%.**

***Index Terms*—machine learning, traffic accident prediction,driving simulation**

I. INTRODUCTION

The Association for Safe International Road Travel (ASIRT) indicates that nearly 1.25 million people die internationally every year from car accidents, and up to 50 million people are injured [1]. Additionally, a decade trend of fatalities in Michigan has shown an upward direction starting from the year 2012. Furthermore, the number of fatalities in 2016 was the highest in the decade after the year 2007. According to Michigan Traffic Crash in 2017, around 314,921 traffic accidents were recorded, which contributed to 1,028 fatalities and 78,394 total injuries. Freeways are one of the locations that are prone to severe form of injury sustained by the involved parties once a crash occurs. Generally, accidents occurrence can be caused by human, roadway, vehicle, and environmental factors. However, the unique characteristics of freeways, including high-speed limits and high traffic volume exacerbate the traffic accidents occurrences. Therefore, the predictive traffic accidents models are vital to make smart decisions that led to avoid accidents on freeways. With the advancements of information technology, machine learning becomes increasingly mature, and useful information without preconditions can be found in databases. Crashes that occur on the highway comprise of various interactions: human, environment, traffic, and roadway-related factors. As a result, the data collected on the causes of highway collisions is very complicated. Improper evaluation may lead to biased recommendations. As a result, it requires a combination of several statistical approaches to address this unobserved heterogeneity. (Mannering, 2016). This study discusses some of the factors that contribute to the occurrence of a crash such as weather condition, road condition and friction, geometry design and speed, and human error.

Supervised classification techniques are popular machine learning methods that aim to explain the dependent variable in terms of the independent variables. Several machine learning techniques have been effectively used to extract useful knowledge from large data sets containing information about traffic accidents. Classification methods are among the most commonly used techniques in mining traffic accidents, where the goal is building classifiers that can predict the disasters. The objective of this study is to identify the root causes of road traffic accidents to apply different machine learning classification techniques for predicting freeway traffic accidents. Then select the most accurate predictive model that will help to reduce these freeway accidents.

II. RELATED WORK

The accidents have worldwide increased over the previous periods and the costs of injuries due to traffic accidents have a significant impact on society. In current years, much studied has been paid attention to find the factors that considerably affect the severity of traffic accidents, and numerous methodologies have been used to study this problem. [2]. The features that are related to traffic accidents include; environmental [3],

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type of vehicle its safety [4], and the characteristics of traffic users. Moreover, some of these factors are more important in determining the accident severity than others. Therefore, the analysis of the determinant factors of accident severity will help reveal more patterns and knowledge that can be used in the prevention of accidents [5]. The prognostic traffic accidents models are reflected to be vital to making smart decisions that can lead to avoiding accidents on freeways. [6] Previous studies exposed that environmental features (i.e., weather condition, roadway surface condition, light condition, crash time influenced the severity of crashes. Numerous studies have considered machine learning methods in transportation-related traffic.Almamlook et al. [7], studied the efficacy of the six machine learning algorithms to build classifiers that are accurate and reliable. In their study, Random Forest (RF), Logistic Regression (LR), Nave Bayesian Classifier (NB), Decision Tree(DT),k-nearest neighbors algorithm (k-NN) and support vector machine. According to the comparison results, the RF algorithm has shown better performance with 75.5% accuracy. They recommended that Random forest is the best model .

Machine learning methods are an achievement, increasing attracting a lot of attention in the arena of transportation. They have exposed their abilities to successfully compact with large numbers of variables while creating powerful predictive models. They also embed variable selection mechanisms which can detect complex relationships in the data. Several studies,suchas [8], [9], [10], have examined machine learning procedures in transportation-related applications of the reasons of accidents.Bahiru et al. [9] compare the accuracy of J48, ID3, CART, and Naive Bayes classifiers in forecasting the severity of injury in traffic accidents. They conclude that the J48 achieved the best with up to 96% accuracy. At the data preprocessing phase, year of accident was considered an irrelevant factor, and it was eliminated. Krishnaveni and Hemalatha [11] showed a prospective analysis of traffic accident events in Hong Kong. In their study, Naive Bayes, AdaBoostM1, J48, PART, and Random Forest classifiers were employed to forecast the severity of injury and reasons for accidents. The Random Forest classifier outperformed all other algorithms. There is no percentage of results.Nakano et al. [12] study the prediction of accident risks and drivers performance among elderly drivers in Japan that are at least 70 years old. They use a logistic regression (LR) study and Support Vector Machine (SVM). Their experiments are shown based on the license renewal test data from the National Police Agency of Japan. This test dataset consists of a driving simulator test and an on-road test. Authors state that non-linear analysis logistic regression classifies most of the cognitively impaired drivers. Moreover, the authors indicate that unimpaired drivers have a strong tendency of becoming impaired and can be predicted earlier. Moreover, they utilized the PART algorithm to present the knowledge in the form of rules, with the accuracy of 79.94% using tool [13].

Therefore, there is a need to perform a comprehensive analysis that aims to understand the relationship between the effect factors and traffic crash outcomes. This study utilized the stimulating driving dataset and used tools to gain better vision and achieve more accurate results. Consequently, this study focuses on using machine learning techniques based driving simulation to predict the Freeway Traffic Accident under difference conduction weather in Michigan, USA. One of the main contributions of this study is to preprocess the Michigan freeway traffic accident based simulation to improve the performance result of the prediction. The study can lead to a greater understanding of the relationship between the weathers condition, vehicle, roadway, and environment and crash. Despite all the approaches realistic to decrease traffic accidents, fatalities and injuries, the severity of the accidents on I69 Michigan freeways is still a real problem that needs paying more attention, studying the environment and extent of the problem

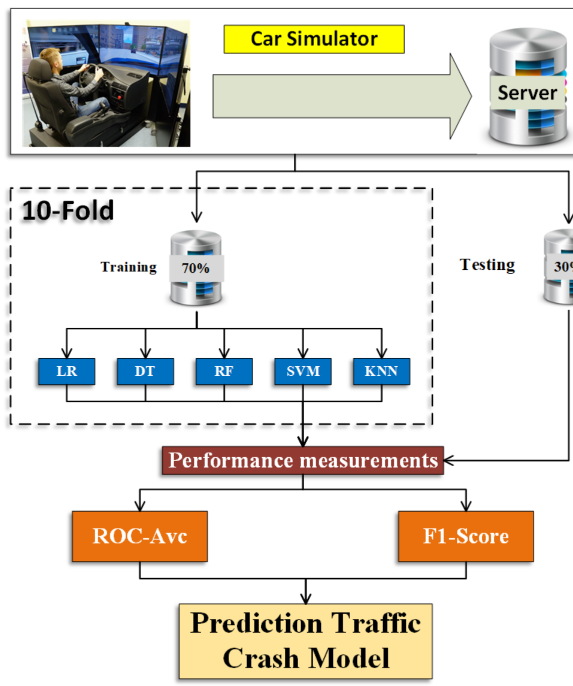

Fig. 1. Methodology block diagram

III. Methodology

TThis study aims to explore the performance of several machine learning techniques in predicting the freeway traffic accidents. As well as identify the main factors that contribute to these accidents. The research methodology can be divided into four steps, including data collection, data pre-processing, building classification models and validating classification models, and knowledge extraction. The classification algorithms have been shown in figure 1 and have explained in the following subsections.

A. Phase I:Study Population and Data description

The data in this study was collected using a driving simulator. The I-69 freeway segment was modeled in the driving simulator in Lawrence University. The simulations involved 100 participants that drove through different simulated roadway scenarios. The scenarios were mainly defined by weather condition and a speed limit of the road. As shown in figure 2, the driving simulator includes a seat, a simulator's computer, a steering wheel, an accelerator, and brakes. The computer's screen displays information for the driver, such as speed in miles per hour (MPH) and revolutions per minute (RPM) and displays the driving scenario. The simulations based on 980 simulated traffic accidents. Also, the simulation involved 100 participants that have driven through ten simulated roadway scenarios according to weather condition with speed limit .

Fig. 2. Methodology block diagram

B. Phase II: Data Prepossessing

This study was proven by the Lawrence Technological University Institutional Review Board (IRB). The data with 18 attributes were collected for the study. Since our research was studied risk factor that causes in the accident-based environment. As for the variable or feature selection, six attributes were selected as possible predictors of crash instances. One hundred subjects (81 males and 19 females) were recruited from the Lawrence Technological University (LTU) community to participate in this driver simulator study. Resulting data were analyzed. From this analysis, a predictive model that focused on driving behavior was developed. There was a total of 1000 observations in our dataset. For our response variables,15% were instances where a crash occurred, and the remaining 85% percent represent situations were crashes didn't happen. The final list of the attributes is presented in Table 1.

TABLE I
SELECTED FEATURES

Feature	Type	Description
$Crash$	C	crash or not
$Speedlimit$	N	speed limit of the freeway
$Coefficientoffriction$	N	Coefficient of friction
$Drivingspeed$	C	Drivers speed
$AHighspeedindicator$	C	drivers exceeded the speed limit or not
$Break$	C	break or no break
$Weathercondition$	C	Weather condition

C. Phase III: Building Classification Modelling

Six well-known classification algorithms were employed to classify the traffic crashes and as well as analysis statistics [14]. These methods are Bayesian Logistic Regression (BLR), Classification tree(DT), Random Forest(RF), (KNN), naive Bayes classifiers (NBC) and Support Vector Machine (SVM). The effectiveness of each method in predicting freeway traffic crashes was evaluated in three different ways. The whole data set was used as a training set for the algorithm, and the accuracy of the classifier was determined based on how well it predicted the class of each accident.

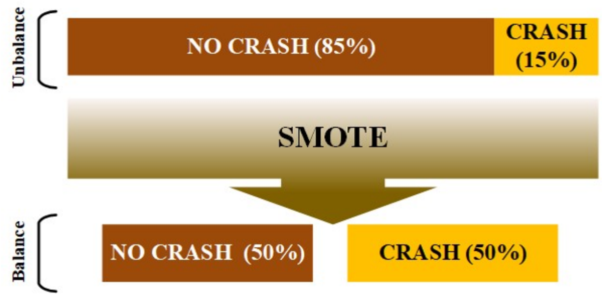

Fig. 3. Conception of unbalanced data

D. Phase IV: Validation Using the Test Set

There are some steps that are used in our experimental. The first step was spitted into training and test set as follows: Training dataset = 70% of the total dataset and Testing dataset = 30% of the total dataset. Second, accuracy was evaluated using cross-validation with 10-fold. The whole data set was randomly partitioned into ten subsets. Of the ten subsets, a single subset was used as the testing data, and the remaining nine subsets were used as training data. As for our case, the data was fairly unbalanced (15% - A crash instance versus 85% - No crash instance) and therefore, we used re-sampling strategy. Model fit using a re-sampling strategy (SMOTE), as shown in figure 3. The algorithm generates synthetic positive instances to increase the proportion of minority class [15]. This process was then repeated ten times, with each of the ten subsets used exactly once as the testing data, resulting in the whole data set being used for validation. Finally, the overall performance is calculated by averaging the ten results from the folds.

IV. RESULTS AND DISCUSSION

Accuracy on the training dataset was used as the criteria for selecting optimal tuning parameters for each model. These parameters were retained in the final model and were used in the test data set for model evaluation. To Measure the predictive performance of the model on the test dataset by comparing Accuracy, Precision, Recall, and F1 Score. Comparison Between Logistic Regression and Other Classifiers: To compare the Logistic Regression classifiers with other typical model in crash accident mode detection from simulation driver data, SIX representative classifiers, i.e., the Classification tree

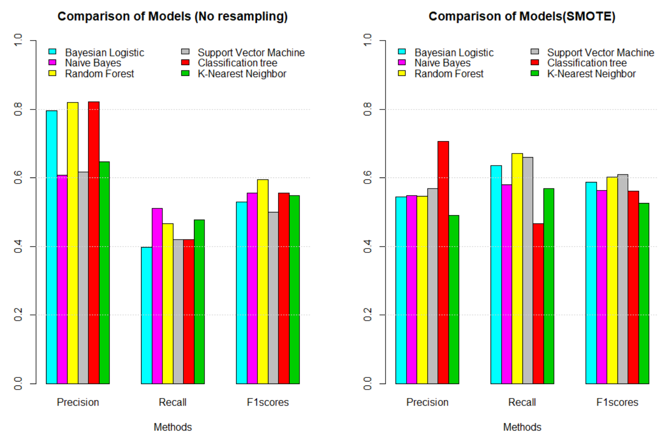

Fig. 4. Model comparison using confusion matrix performance measures

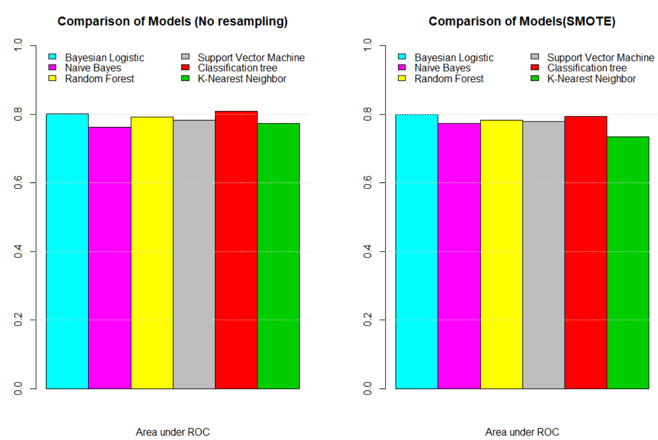

Fig. 5. : Model comparison using ROC curve

(DT), naive Bayes classifiers, Random Forest(RF), k-nearest neighbors algorithm (k-NN), Bayesian Logistic Regression (BLR), and Support Vector Machine (SVM) are employed to the same selected feature set. The sample is also split into 30% and 70% for model training and testing, respectively. Overall for all models, the recall value was improved when applying SMOTE. However, there was a decrease in the precision of the model of using SMOTE. Based on the confusion matrix F1-score, Random Forest seemed to perform better than the other models.Based on area under ROC curve, Classification Tree, Bayesian Logistic regression and Random forest had an area of about 0.8.After an accurate comparison between our algorithms, we noticed that RF achieved a higher efficiency of 0.82.Two scenarios were compared one without applying any re sampling strategy and one by applying the re sampling strategy. Overall for all models the recall value was improved when applying SMOTE. However, there was a decrease in precision of the model of applying SMOTE. No significant difference where observed two scenarios were compared using area under ROC curve. The increase in recall value after applying are sampling strategy demonstrates the usefulness of applying SMOTE technique in a dataset that have under representation of relevant cases that a researcher wishes to identify.

As shown in Figure 4, the Random Forest classifier achieves the highest accuracy among all the classifiers for the testing set with Smote. Therefore, there is no doubt that the Random Forest classifier has an advantage over the other three representative classifiers in this sample. This result demonstrates the best ML technique is Random forest. Compared with different classifiers, Random Forest has an excellent ability to resist noise due to the application of randomly selecting variables and data to generate plenty of classification trees. It can process not only discrete data but also continuous data. The result proves that it is adaptable for diverse and high-dimensional data sets, which enables the RF classifier in mode detection incorporating multi-source attributes.

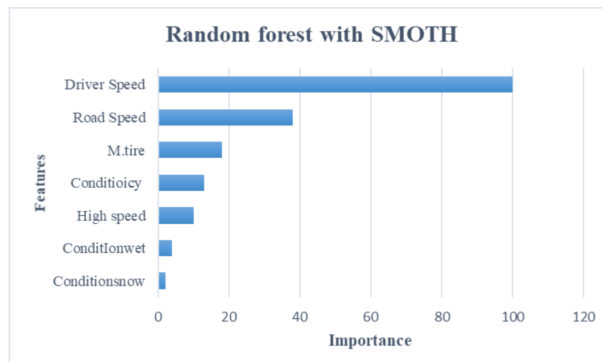

Fig. 6. :Feature important of Random forest with SMOTE

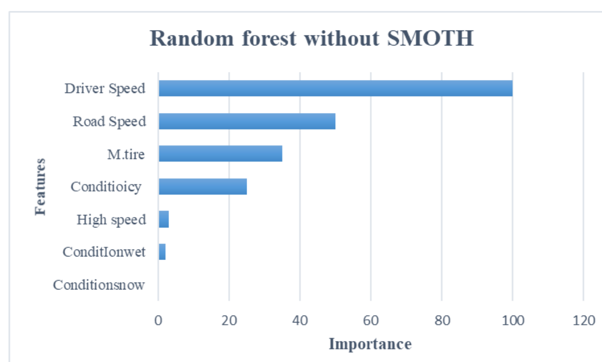

Fig. 7. : Feature important of Random forest without SMOTE

Generally, AUC ranges from 0.5 to 1, and the higher the AUC is, the better the classification performance is. When the AUC is higher than 0.8, it indicates that the model has an excellent prediction ability in pattern recognition. The AUCs with SMOTE ,six modes detection in our study are 0.79, 0.81, 0.82,0.61,0.60, and 0.65 for LR, DT, RF, SVM, NBC, and KNN, respectively as shown in Fig 4. These encouraging (AUC) give statistical proof to the excellent classification capacity of Random Forest in this study. To better understand efficiency, Figure 5 presents the ROC curve of our classifiers that better illustrate the precision of each classifier. The ROC curve gives a graphical graph that illustrates the performance of different classifiers. From the plot, we can easily select optimal models and discard others to the best classification. According to a random forest with SMOTE and without SMOTE, speed is the most critical factor to predict the injury severity. Road speed is the second factor, while the M.tire is the third. Figs. (6 and 7) show the feature importance to both with SMOTE and without SMOTE.

V. Summary and Conclusions

The test result shows that based on the confusion matrix F1-score, Random Forest seemed to perform better than the other models. Based on the area under the ROC curve, Random forest had an area of about 0.826. We recommend Random forest as our best model for predicting freeway crashes.Also, speed is the most critical factor to predict the injury severity.The recommended model can be applied to monitor freeway crashes. The recommended predictive model can be used to rapidly and efficiently identify the critical factor causing traffic crashes. The dataset used in this study was simulated crash data. More realistic results can be obtained using the actual accident data model with the test data.

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